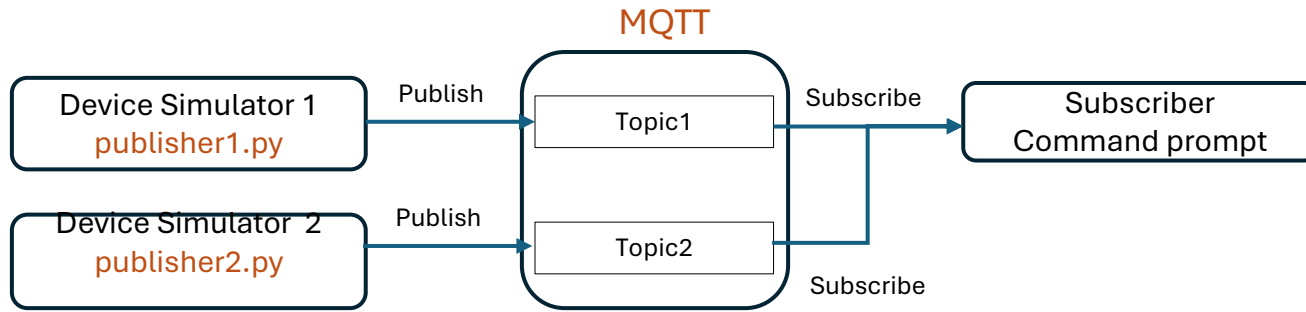


1. Assignment – Communicating with MQTT



Empowering Patient Care through Innovative Technology

Create two device simulators to enhance hospital patient monitoring.

- Device #1 will publish heart rate and blood pressure every minute.
- Device #2 will monitor room temperature every 30 seconds.

Using MQTT, connect to an external server to observe real-time data.

Empowering Patient Care through Innovative Technology

Your mission is to create 2 device simulators, which will ensure that both patients and their environments are carefully monitored for optimal care within a hospital setting.

Project Overview:

1. Device Simulators:

1. **Device #1:** This medical equipment will be responsible for publishing crucial patient parameters, such as heart rate and blood pressure, every minute. This real-time data will empower healthcare providers to make informed decisions swiftly.
2. **Device #2:** This device will monitor the temperature in patient rooms, publishing updates every 30 seconds. Maintaining the right environment is essential for patient comfort and recovery.

2. Technical Setup:

1. Utilize MQTT for communication, ensuring the hostname is set to connect with an external server (not localhost).
2. You can either specify a dedicated hostname for your MQTT connection or leverage a public message broker available online, such as [HiveMQ](#).

3. Message Monitoring:

1. Once your devices are operational, you'll view the messages published by both devices directly in your local command prompt. This step will allow you to witness the impact of your work in real-time.

Inspiration: Imagine the difference your project will make in patient care! By building these simulators, you are contributing to a future where technology and healthcare intersect to save lives and improve well-being.